Heat Transfer to a Fluid in a Stirred Tank

Introduction

The agitated tank is used for a variety of different processes, for example, chemical reactions, blending, dispersions and leaching. A typical system consists of the stirred tank (usually assumed to be perfectly mixed, or at constant bulk properties), and either a steam jacket to provide heating or an internal cooling coil. Baffles are often used to enhance mixing. Additional vessel fluid cooling can be achieved by a recirculation circuit in which the fluid is passed through an external cooler.

This apparatus presents the opportunity to study a range of heat transfer processes. Of main concern here will be the measurement of the heat transfer coefficient between the fluid and inside vessel wall and to examine the dependence of this heat transfer coefficient on the impeller speed, fluid properties (which will vary with temperature), and the presence of baffles. The experimentally measured heat transfer coefficent will also be compared to values reported by other investigators.

The values of the heat transfer coefficients are functions of the fluid flow field and the molecular transport properties of the fluid. In many instances, as for example the agitated vessel, the flow pattern is too complex to be analyzed analytically. As a result, heat transfer rates must be determined experimentally. However, experimental data, theoretical analysis, and dimensional analysis have shown that the heat transfer coefficient between a fluid and a surface can frequently be correlated in terms of a relatively small number of dimensionless groups. For convective flow (in general, the rapid movement of a fluid over a surface provides an effective means of heat transfer between the fluid and the surface), usually only the Reynolds number (Re), Prandtl number (Pr), and geometric factors are needed to correlate heat transfer data over a broad range of conditions. Thus, heat transfer coefficients are often reported as functions of these dimensionless groups. Correlations have been formulated for design purposes for many standard cases. These can be found in an engineering handbook such as Perry's.

Experimental data for the inner wall heat transfer coefficient for an unbaffled vessel containing a Newtonian fluid, have been summarized by the following correlation:

$$\frac{h_i D_T}{k} = 0.36\left(\frac{D_I^2\, n\rho}{\mu}\right)^{\frac{2}{3}}\left(\frac{C_p\,\mu}{k}\right)^{\frac{1}{3}} \qquad (1)$$

where

- h_i = convective heat transfer coefficient at the inner surface of the tank

- D_T = tank diameter
- μ = viscosity of the fluid
- C_P = specific heat of the fluid
- k = thermal conductivity of the fluid
- ρ= density of the fluid
- D_I = impeller diameter
- n = speed of rotation of the impeller, (revolutions per unit time)

The heat transfer coefficient is defined for the difference in temperature between the inside vessel wall and the bulk mean temperature of the well stirred fluid. The fluid properties are those of the liquid in the tank at the bulk mean temperature of the fluid.

The dimensionless groups for this correlation are:

$$\frac{hD_T}{k} \equiv Nu_T = \text{tank Nusselt number}$$

$$\frac{D_I^2 n \rho}{\mu} \equiv Re_{imp} = \text{impeller Reynolds number}$$

$$\frac{C_p \mu}{k} \equiv Pr = \text{fluid Prandtl number}$$

In terms of these dimensionless groups, equation (1) may be written as:

$$Nu_T = 0.36 \{Re_{imp}\}^{2/3} \{Pr\}^{1/3} \qquad (2)$$

Since these groups are dimensionless, any consistent set of units may be used in their evaluation. Note that the Reynolds, Prandtl, and Nusselt numbers are standard dimensionless groups which usually always appear in correlations for heat transfer coefficients; however, the characteristic length scales with respect to which these groups are defined will depend on the geometric characteristics of the system. (For example, in this correlation the Reynolds number is defined in terms of the impeller diameter and the impeller tip speed.) The fluid properties used to evaluate these groups corresponds to that fluid through which heat transport is taking place. It is important that the length scales and fluid properties be clearly defined in any correlation and care should be used that these groups are correctly evaluated in a specific correlation .

The heat transfer coefficient on the outside of the vessel wall, where film condensation from the steam jacket occurs, can be estimated from the correlation.

$$\frac{h_o L_T}{k} = 0.925 \left(\frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{\frac{1}{3}}, \qquad (3)$$

which is valid for $Re_{film} < 2100$, where $Re_{film} = (4\Gamma/\mu)$, and where

- Γ = mass rate of steam condensation per wetted perimeter

$$= \frac{\dot{m}_{st}}{\pi D_{T,o}}$$

- $\dot{m}_{st}$ = the mass rate of steam condensation
- $D_{T,o}$ = the outer diameter of the vessel wall
- h_o= the HTC through the condensing film
- L_T = vertical length of the tank
- g = acceleration due to gravity
- μ = the viscosity of the condensed steam.
- ρ = the density of the condensed steam

The heat transfer coefficient is defined for the temperature difference between the outer wall temperature and the saturation temperature of the steam.

For saturated steam at 1 atm pressure, h_o evaluated from this equation is equal to:

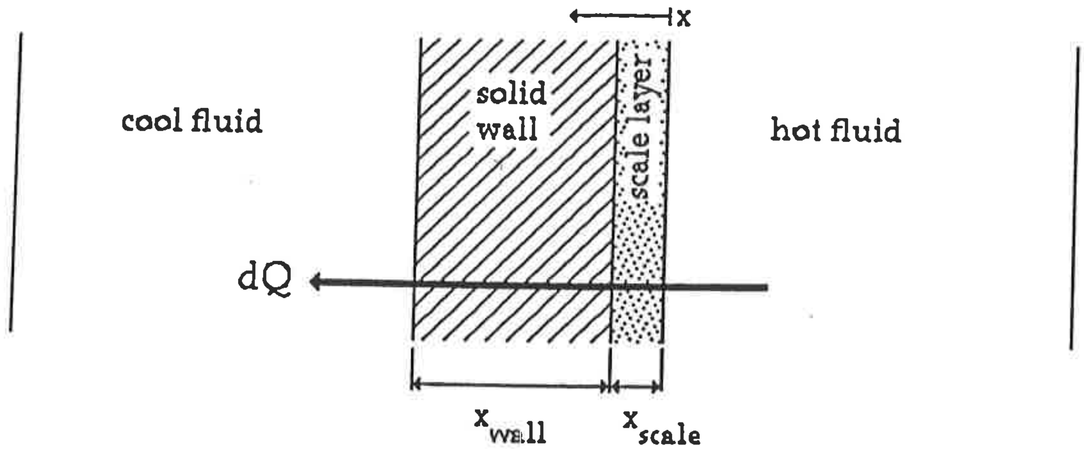
$$h_o = 2960(D_{T,o}/m_{st})^{1/3} \tag{4}$$

where h_o is the steam side heat transfer coefficient in W/m^2-K; $D_{T,o}$ is the outer tank diameter in m; and m_{st} is the mass flowrate of steam in kg/s. This equation is dimensional, and the above units must be used for evaluating h_o.

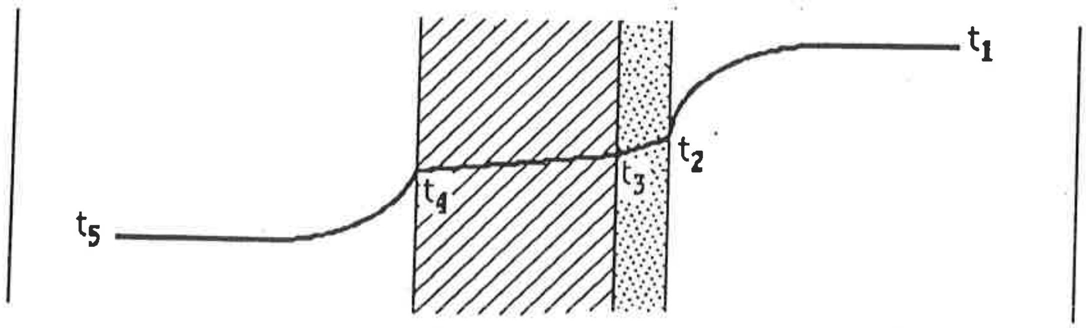
No general correlations exist for the case of agitation with baffles in a jacketed tank. The presence of the baffles deters the formation of a vortex, which decreases the contact area between the fluid and wall. However, the flow profile has more of a radial component, and the fluid is better mixed. Also, baffles permit the use of higher impeller speeds. The net effect is that in a fully baffled vessel, the heat transfer rate is many times higher than for an unbaffled vessel.

Theory

A microscopic view of heat transfer from one fluid to another across a solid wall reveals a number of processes in series. In the schematic below it is assumed that heat transfers from right to left, and that a layer of scale (mineral deposits) has formed on the hot side of the wall. (In some cases it may not form at all or form on the other side.) Heat must transfer across the hot fluid-scale interface, be conducted across the scale film and the solid wall, and finally must cross the cool side wall-fluid interface. At steady state, the heat transferred, Q, is the same at any point in the x direction.



Each interface and solid phase presents a different resistance to heat transfer, such that the temperature profile at steady state is established as follows:



Q can be expressed for the two liquid-solid interfaces as

$$Q = h_i\, A_i\, (t_4 - t_5) = h_o\, A_o\, (t_1 - t_2), \tag{5}$$

where h_i is the film heat transfer coefficients at the inner surface, which has an area A_i, and h_o is the heat transfer coefficient at the outer surface, which has an area A_o. t_1 is the temperature of the hot fluid; t_2 is the temperature of the inner wall; t_4 is the temperature of the outer wall; and t_5 is the temperature of the cool fluid. Note that the heat flux, Q, through each individual resistance must be the same and is equal to the total heat flux across the interface. At steady state, there can be no accumulation of heat, and the heat flux across the interface is a constant, as shown in the first figure.

Heat transport through the two solid surfaces can be expressed by Fourier's law:

$$Q = k\,A\,\frac{dt}{dx}$$

$$= A_{scale}\, k_{scale} \frac{(t_2 - t_3)}{x_{scale}} = A_{wall}\, k_{wall} \frac{(t_3 - t_4)}{x_{wall}} \tag{6}$$

where k_{scale} and k_{wall} and x_{scale} and x_{wall} are the thermal conductivities and thicknesses, respectively, of each solid material.

The above equations for heat transfer across each individual resistance in terms of the local interface temperatures can be combined and expressed in terms of the temperature difference, $t_1 - t_5$, across the composite interface and an overall heat transfer coefficient, U_i. This is useful since the individual interface temperatures may be difficult or impossible to measure directly. If the area considered is the inner area of the wall, then

$$Q = U_i A_i (t_1 - t_5). \tag{7}$$

and

$$Q = U_i A_i [(t_1 - t_2) + (t_2 - t_3) + (t_3 - t_4) + (t_4 - t_5)], \tag{8}$$

Now each local temperature difference may be expressed in terms of the total heat flux, Q, and the local heat transfer coefficient or the conductivity and thickness of each solid resistance as follows:

$$(t_1 - t_2) = \frac{Q}{h_o A_o}, (t_2 - t_3) = \frac{Q x_{scale}}{A_{scale} k_{scale}}, \text{ etc.}$$

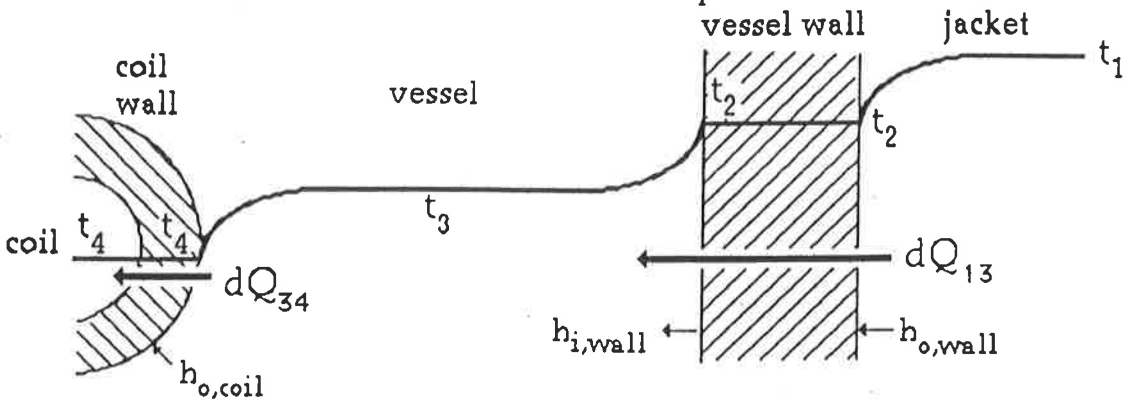
After substituting and simplifying, it is found that

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{x_{scale}}{k_{scale}} \frac{A_i}{A_{scale}} + \frac{x_{wall}}{k_{wall}} \frac{A_i}{A_{wall}} + \frac{1}{h_o} \frac{A_i}{A_o}}, \text{ or} \tag{9}$$

$$\frac{1}{U_i} = \frac{1}{h_i} + \frac{x_{scale}}{k_{scale}} \frac{A_i}{A_{scale}} + \frac{x_{wall}}{k_{wall}} \frac{A_i}{A_{wall}} + \frac{1}{h_o} \frac{A_i}{A_o} \tag{10}$$

This expression is analogous to the calculation of the overall resistance for resistors in series.

For the purposes of this experiment, calculations for the overall heat transfer coefficients of the wall will be based on the simplified scenario shown below:



The simplifying assumptions are:

- no scale on either side of either solid wall and

- negligible resistance to heat transfer by the solid wall.
  That is, the temperature gradient across the solid wall is small. This will occur if the thermal conductivity of the solid is high, (as would be the case for many metals) and the wall thickness is small

Calculation of the overall heat transfer coefficient will further include the simplification that the wall thickness is small compared to the tank diameter, such that $A_i/A_o \approx 1$. Then,

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{1}{h_o}} \tag{11}$$

and

$$Q_{13} = U_{i,wall}\,(t_1 - t_3)\,A \tag{12}$$

Here t_1 is the steam temperature in the jacket, t_3 is the bulk average temperature of the fluid in the tank, and A is the area of the heat transfer surface. The liquid in the vessel can be assumed to be well mixed and thus at a uniform temperature.

Apparatus and General Procedure

A schematic of the jacketed, agitated vessel with recirculation stream and internal cooling coil is shown in the following figure. Dial thermometers are provided at the inlet and the outlet of the coil. Cooling water enters the coil through a rotameter which indicates the water flowrate. Water from the tank is continuously withdrawn from the bottom of the vessel and circulated through the shell side of a heat exchanger where it is cooled by cold water flowing through the tube bundle. The cooled tank water passes through a rotameter and is returned to the vessel. The temperature of the recirculated tank water is measured both at the inlet and outlet of the heat exchanger. The water temperature inside the tank is also indicated by a dial thermometer. The impeller speed is measured by a [illegible] and can be varied with a variable power supply. tachometer ~~[illegible]~~ (Ask TA for instrument connected to apparatus, or an external unit)

<u>General Operating Procedure</u>:

1) Fill the tank with water to a level about two inches from the top of the tank.
2) Turn on the impeller and adjust the speed to ~~approximately 400 rpm~~. >600 RPM*
3) Turn on the cooling water to the cooling coils; open the control valve fully.
3) Turn on cooling water to tube side of the shell and tube heat exchanger.
4) Make sure that outlet valve in the discharge line of the pump is fully closed (why?) and the suction and bypass valves are fully open.
5) Switch on the pump and allow it to run for a few seconds with discharge valve closed.

* Note: impeller speed must be >600 RPM for more accurate speed control. Impeller can also be run clockwise or counterclockwise.

6) Open the discharge valve slowly and close the bypass valve to set a particular recirculation rate. Make sure that the circulating liquid is fed back to the tank.
7) Slowly open inlet steam valve (perhaps 1 1/2 turns) after turning on cooling water to steam condenser. Later calculations will be simplified if the steam flow rate is low enough so that all steam condenses completely in the jacket. This will be evidenced by only a very small temperature increase in the shell side fluid of the steam condenser. The quality of the steam leaving the jacket can be found in any case using an energy balance around the steam condenser.
8) At steady state, record all relevant properties i.e., flow rates, temperatures, speed of impeller, etc.

Measurement of Heat Transfer Rate

The heat transferred to the vessel from the steam, Q_{13}, can be determined from an energy balance on a control volume around the steam jacket and steam condenser.

$$Q_{13} = \dot{m}_{st}(h_{st,in} - h_{st,out}) - Q_{sc} - Q_{surr} \tag{13}$$

Here, $h_{st,in}$ is the enthalpy of the entering steam and $h_{st,out}$ is the enthalpy of the steam condensate. In order to determine the enthalpy of the entering steam, the quality of the steam must be known. This can be measured using the steam calorimeter that is connected to the steam line. The enthalpy of the condensate must be determined relative to the same reference state used to determine the steam enthalpy. Q_{sc} refers to the heat removed in the steam condenser. This may be measured by measuring the cooling water flowrate to the condenser and the temperature rise of the cooling water. Q_{surr} is an estimate of the heat loss of the insulated vessel to the surroundings.

The heat transferred to the fluid can also be determined by consideration of a heat balance around the agitated vessel since at steady state the amount of heat added to the fluid in the vessel must be equal to the amount of heat removed. Heat is removed from the fluid by either the cooling coils or the external heat exchanger. (This assumes that negligible energy is lost along the recirculation tubing or due to evaporation of water from the tank.) A steady state heat balance on the vessel fluid then gives:

$$\text{Heat in} = \text{Heat out}$$

or:

$$Q_{13} = Q_{hx} + Q_c + Q_{suur'} \tag{14}$$

Here, Q_{hx} is the amount of heat removed in the external heat exchanger, Q_c is the amount of heat removed by the cooling coils, and $Q_{surr'}$ is heat loss from the recirculation pipes and due to evaporation of water from the vessel .

Q_{hx} can be determined from the flowrate and the temperature drop of the fluid recirculated throught the heat exchanger:

$$Q_{hx} = m_{hx}C_p(t_3-t_r) \tag{15}$$

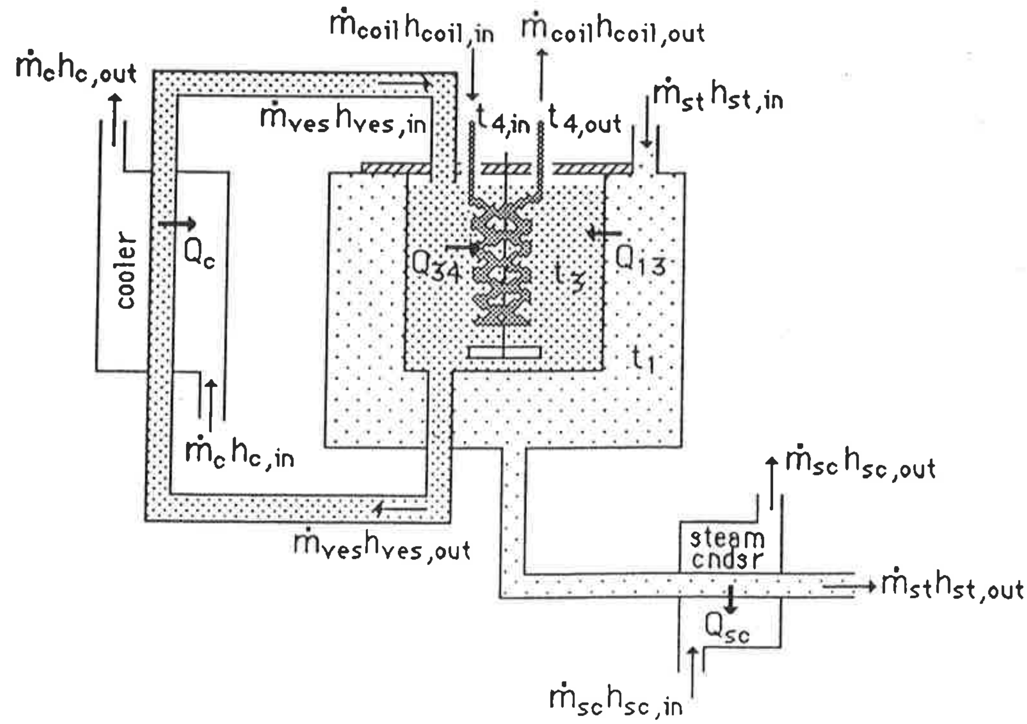
Here, m_{hx} is the amount of fluid circulated through the heat exchanger; C_p is the heat capacity of the fluid (is this quantity a constant?); t_3 is the temperature of the fluid entering the heat exchanger (this temperature should be the same as the vessel temperature if the fluid is well mixed!); and t_r is the temperature of the fluid leaving the heat exchanger and returned to the tank.

The heat removed by the cooling coils can be determined from the flowrate of water in the coils and the temperature rise of the water as it passes through the coils:

$$Q_{coils} = m_c C_p(t_{out} - t_{in}) \tag{16}$$

Here m_c is the flowrate of water through the cooling coils, C_p is the heat capacity of the water, and $(t_{out} - t_{in})$ is the temperature rise of the cooling water in the coils.

If all heat flows are correctly accounted for, Q_{13} determined by either control volume heat balance should give the same number.



Two of the three experiments outlined below will be assigned by the instructor.

Experiment A. Wall HTC for the Unbaffled Agitated Tank

Procedure:

1a) $h_{i,wall}$ ~ 600 RPM

At a constant steam flow rate to the vessel, a low impeller speed ~~(about 150 rpm)~~, with water flowing through the cooling coils, and with the heat exchanger flowrate such that the temperature in the vessel (t_3) is about 65°C (try to keep the vessel temperature below 85°C in most of your experiments.), record the measurements needed to calculate Q_{13} from equations (18) and (19).

1b) $h_{i,wall}$ as a function of impeller speed.

Repeat the above measurements at two additional higher impeller speeds.

1c) $h_{i,wall}$ as a function of temperature.

Repeat the measurements for two different heat exchanger flowrates so that two different vessel temperatures are obtained. Use your second impeller speed settings.

Results and Discussion

- Decide which control volume balance gives the most accurate value for Q_{13} and state why.
- What is the steam temperature in the vessel jacket, if it is assumed that the steam pressure in the vessel jacket is 1 atm.?
- Calculate the overall wall heat transfer coefficient, U, for each case 1a→c, using that value of Q_{13} you believe is more accurate. Calculate the corresponding outer film HTC, $h_{o,wall}$, using the appropriate correlation. Note that when making this calculation, the quality of the steam entering and leaving the vessel jacket must be known in order to calculate the rate of condensation, Γ. Use these correlation values for $h_{o,wall}$ to calculate the experimental $h_{i,wall}$ values from the measured values of U. For each case, also compute $h_{i,wall}$ values from the given correlation. Summarize these calculations in a table.
- Compare your measured $h_{i,\ wall}$ values to those obtained from the correlation. The correlation for $h_{i,wall}$ can be rearranged as:

$$\frac{\left(\frac{hD_T}{k}\right)}{\left(\frac{C_p\mu}{k}\right)^{.33}\left(\frac{\mu}{\mu_w}\right)^{.14}} = 0.36\left(\frac{D_I^2 n\rho}{\mu}\right)^{.67} \quad \text{or} \tag{17}$$

$$\log\left[\frac{Nu_T}{Pr^{1/3}}\right] = \log(0.36) + 0.67\log[\,Re_I\,]; \tag{18}$$

A plot of the left hand side versus $\log\{Re_{imp}\}$ will yield a line of slope .67. Graph a line corresponding to equation (18) over the range of impeller Reynolds number employed in this experiment. The Reynolds number will increase with increasing impeller speed as well as with increasing temperature. Plot your experimental points for $h_{i,wall}$ on the same graph.

Agreement between the measured and correlated values may not be very good; the correlation given in (2) is for a paddle stirrer, whereas in this experiment an impeller was used to agitate the fluid; however, trends should be in general agreement. Try to explain the trends that you observe.

Experiment B
Wall HTC for the Baffled Agitated Tank

Procedure:

Measurements will be made at approximately the same conditions as in Experiment A, except with the baffles inserted. Should Experiment B be assigned, these additional measurements are a simple matter. After steady state has been reached in each case of Experiment A, carefully lift the vessel cover plate and slip the baffles into place, making sure the top flat section is seated over several allen screw heads. After the new steady state has been reached, take all necessary measurements.

Results and Discussion

- Calculate $h_{i,wall}$ as before for each set of conditions. Summarize all calculations in a table. (Will the outer vessel wall HTC , be affected by the baffles?)
- Tabulate experimental and correlation values for $h_{i,wall}$. State trends in the HTC with impeller speed and temperature with the baffles in place. Does the change in the flow pattern with the baffles increase the rate of heat transfer?
- Overlay the baffled case HTCs on the previous plots for $h_{i,wall}$. Does

the data show the same dependence on Re_{imp}? Try to explain the differences obtained in the baffled case.

- Assuming that a new correlation is needed for this case, with Re_{imp} and Pr as dimensionless groups, show how one could obtain a value for the exponent on the Reynolds number and for the constant in the correlation.

Experiment C
Unsteady State Wall HTC for the Unbaffled Agitated Tank

<u>Procedure:</u>
In this experiment the unsteady state heat transfer rate is measured as a function of time; the recirculation pump and the cooling coil will not be used so as to to simplify the experimental procedure and measurements. Initially the steam is shut off, but the cooling water flow to the steam condenser is left on.

At non-steady-state conditions, a heat balance on the fluid in the vessel gives:

Heat in - Heat out = Accumulation of heat in the fluid

In terms of measureable quantities, this equation can be written as:

$$A \cdot U \cdot (t_1 - t_3) - 0 = m_v C_p \{dt_3 / d\Theta\} \qquad (19)$$

where the first term on the left hand side refers to the rate at which heat is added to a well agitated fluid in the vessel. The heat loss terms are assumed to be zero since both the cooling coil is off and no fluid is recirculated through the heat exchanger. The term on the right hand side accounts for the increase in the internal energy of the fluid with time. The symbols m_v refers to the mass of fluid in the vessel, C_p is the heat capacity of the fluid, Θ is time, and $dt_3/d\Theta$ is the change in the fluid temperature with time. A value for $dt_3/d\Theta$ can be obtained by measuring the slope of a graph of t_3 vs. Θ at appropriate temperatures. The overall heat transfer coefficient, U, can then be calculated from the equation:

$$U = \frac{m_v C_p \frac{\Delta t_3}{\Delta \Theta}}{A(t_1 - t_3)} \qquad (20)$$

where $\Delta t_3 / \Delta\Theta$ is the derivative of t_3 vs Θ at temperature t_3.

3a) Starting with the median impeller speed used in Experiment A, and at a low temperature (≤ 40°C), open the steam valve a few turns. Carefully record the fluid temperature as a function of time. (how good is your assumption that the fluid is well mixed?) Record data until a temperature of about 85°C is reached. Ten to twenty data points should be recorded. Two sets of measurements should be obtained. The fluid in the vessel can be cooled quickly by turning off the steam and recirculating water through the heat exchanger and cooling coils.

3b) Repeat the experiment using one additional impeller speed.

Results and Discussion

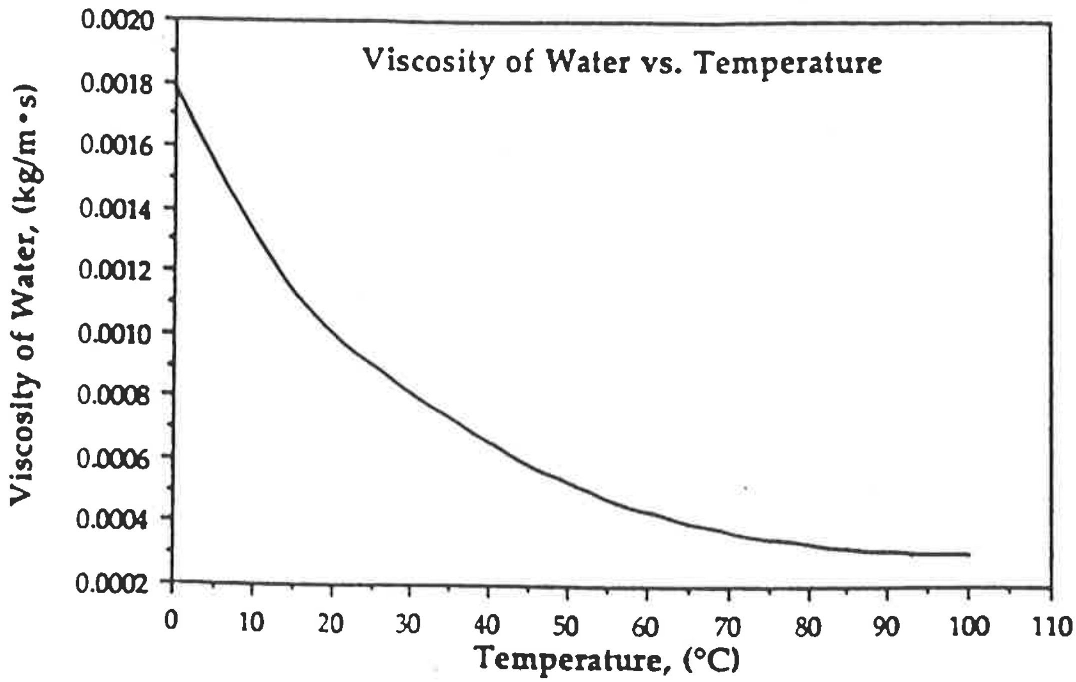
- Using $\left(\frac{\Delta t_3}{\Delta\theta}\right)$ to approximate $\left(\frac{dt_3}{d\theta}\right)$, calculate Q_{13}, $U_{i,wall}$, and $h_{i,wall}$ as functions of time and temperature; $h_{o,wall}$ must be estimated from correlation (3) as before;.

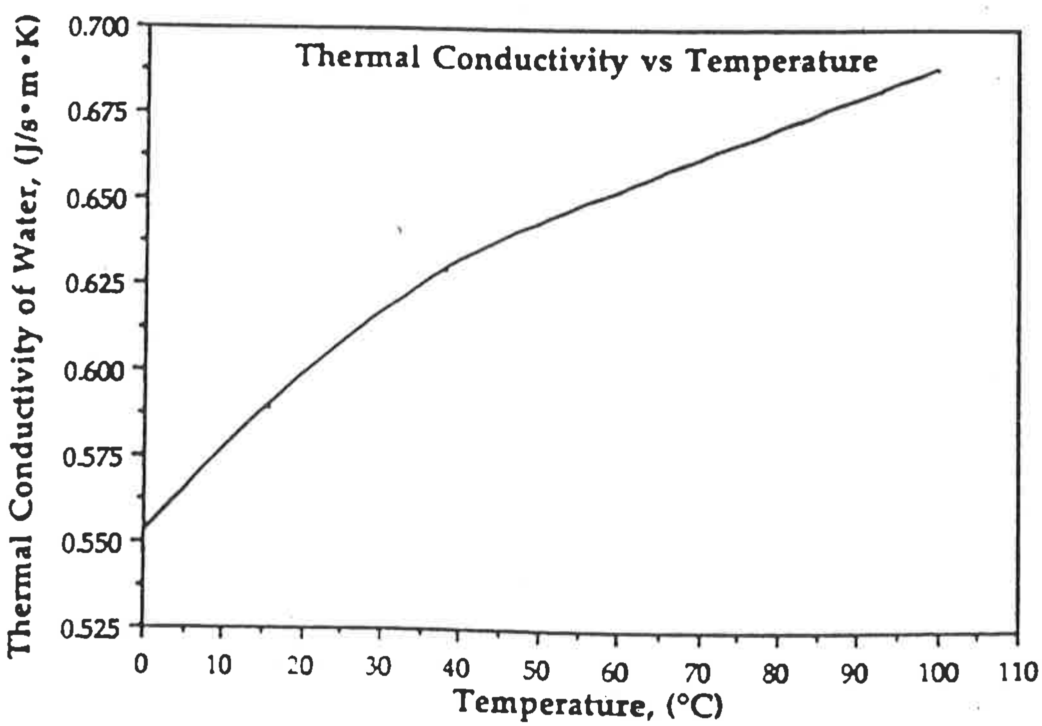
- Plot $h_{i,wall}$ on the correlation plot developed in the previous part of this experiment. Both the Reynolds and Prandtl numbers will take on different values at different temperatures, t_3

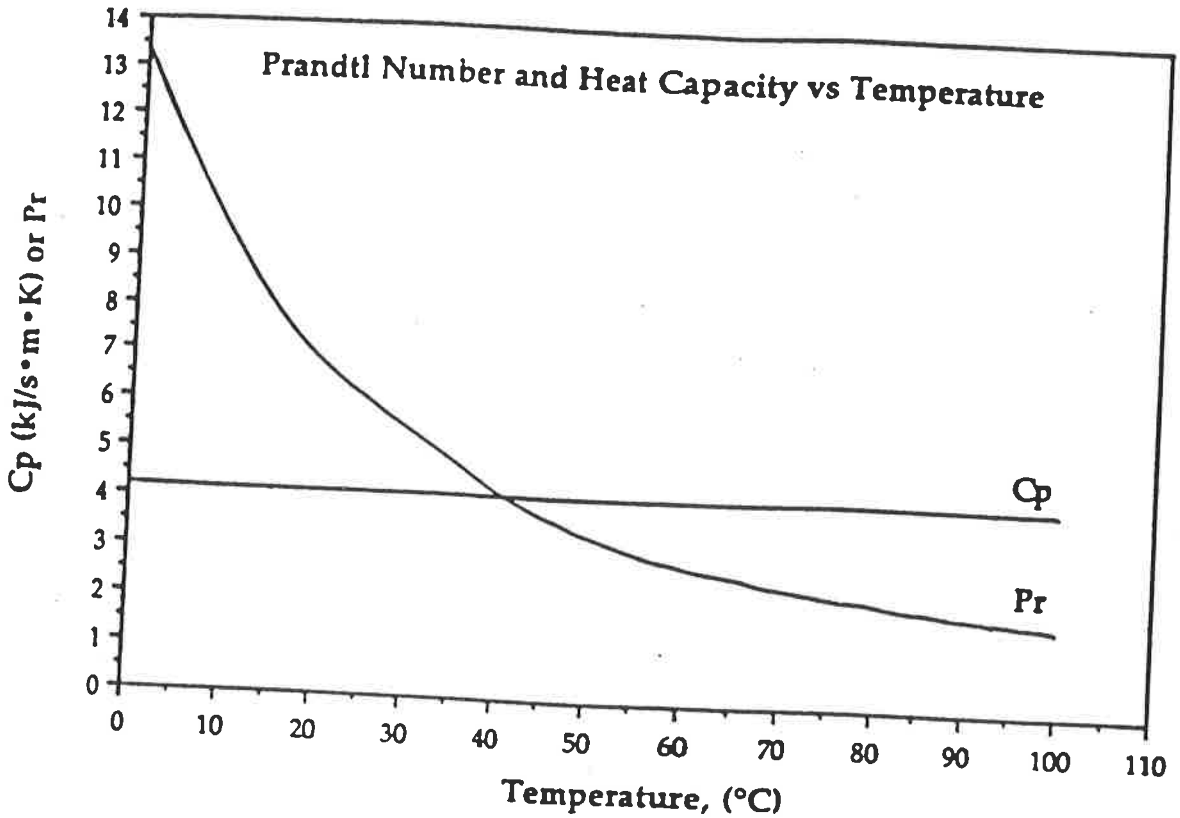
List of Symbols SI (units)

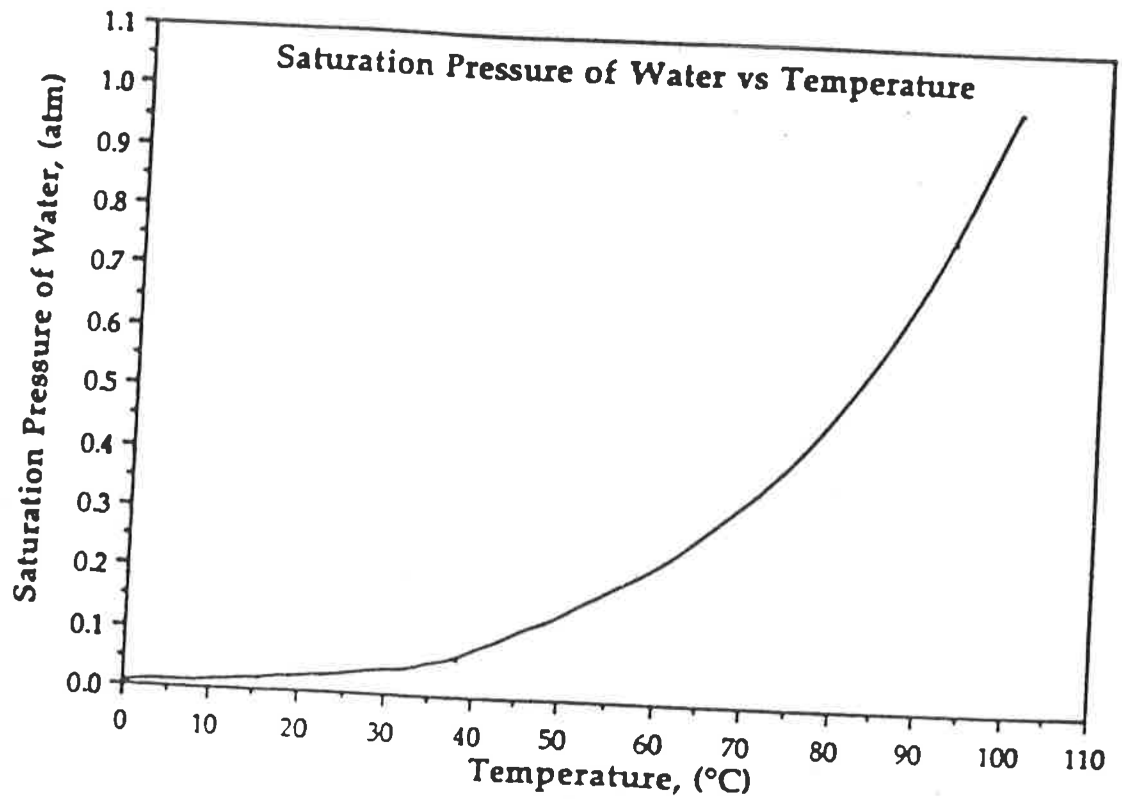
Q_{13}	heat transferred from steam to fluid in vessel (W)
Q_{hx}	heat removed by external heat exchanger (W)
Q_c	heat removed in cooling coils (W)
Q_{surr}	miscellaneous heat loss to surroundings (W)
h_i; $h_{i,wall}$	heat transfer coefficient at innner vend wall (W/m^2 - K)
h_o; $h_{o,wall}$	heat transfer coefficent at outer wall surface (steam side) (W/m^2 - K
m_{st}	steam flowrate (kg/s)
m_{hx}	recirculation rate (kg/s)
m_c	cooling water flowrate in coils (kg/s)
m_v	total fluid in vessel (kg)
D_I	impeller diameter (m)
D_T	inner tank diamter (m)
$D_{T,o}$	outer tank diameter (m)
L_T	vertical length of tank (m)
A	heat transfer surface area (m^2)

Appendix A. Properties of liquid water at its saturation state as a function of temperature.






Prandtl Number and Heat Capacity vs Temperature
Cp (kJ/s•m•K) or Pr
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
Cp
Pr
0
10
20
30
40
50
60
70
80
90
100
110
Temperature, (°C)


Saturation Pressure of Water vs Temperature
Saturation Pressure of Water, (atm)
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
10
20
30
40
50
60
70
80
90
100
110
Temperature, (°C)

ChE 382: Unit Operations Laboratory

HEAT TRANSFER TO A FLUID IN A STIRRED TANK

rev: Fall 2005

Introduction

Stirred tanks are widely used in the chemical processing industry. They come in a great variety of sizes, materials of construction, and operational parameters such as stirring and internal baffles. Examples of use include simple mixing of solutions, leaching, and chemical reactions.

Heat transfer in these systems can be an important factor and due to the great variety of types of stirred tanks there are few reliable correlations. It can be necessary to add heat to a system to avoid phase changes or cool the system in the case of an exothermic polymerization reaction. In many cases, designing for heat transfer can be crucial for successful use of this unit operation. The stirred tank in the Unit Operations Laboratory will be used to study a range of heat transfer processes involving the basic stirred tank unit. The focus will be on the measurement of the heat transfer coefficient between the fluid and the inside vessel wall. Parametric studies will include the dependence of the heat transfer coefficient on the impeller speed, fluid properties, and the use of mechanical baffles.

Theory

The heat transfer process consists of three basic steps:

- Heat transfer across the internal fluid to the wall of the stirred tank
- Heat transfer across the tank wall
- Heat transfer from the condensing steam to the tank wall

The heat transfer coefficient U is usually written in terms of the temperature driving force from the steam to the internal fluid:

$$Q_{13} = U(t_1 - t_3)A \tag{1}$$

The overall heat transfer coefficient describes the total resistance of the three individual resistances described above. The references include a discussion of how this can be calculated from the individual heat-transfer coefficient.

Some simplification of the analysis can be made for this particular experimental unit. First, the wall thickness is small compared to the tank diameter and thus the area for the heat transfer is essentially the same on both sides of the wall. Second, the tank is constructed of a high-conductivity stainless steel and its resistance can be assumed small compared to the other two resistances. These assumptions lead to a simple expression for the overall heat transfer coefficient:

$$U = \frac{1}{\frac{1}{h_i} + \frac{1}{h_0}} \tag{2}$$

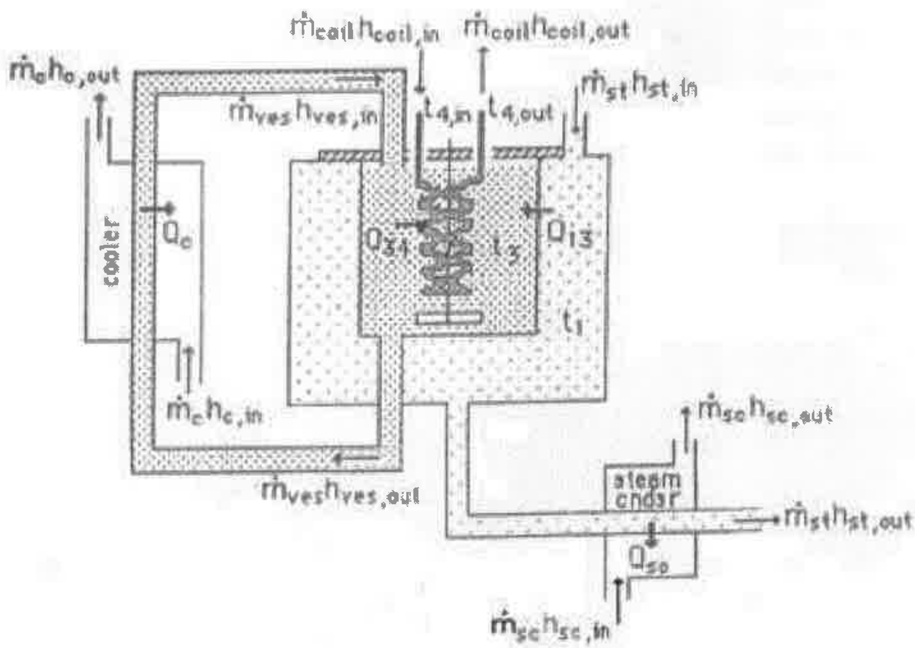
Where h_i is the heat transfer coefficient from the fluid to the tank and h_o, the heat transfer coefficient from the wall to the condensing steam.

There are available correlations for the heat transfer from condensing steam to the tank wall. They tend to be very large since this is a highly turbulent process with rapid exchange between the steam and the surface. These correlations can be found in Perrys [2] or the other suggested reference.

There are, however, no general correlations for heat transfer coefficients in baffled, and stirred tanks. Thus, experimental values are often required for rational process design.

Experimental

A schematic diagram of the stirred tank unit is shown below:



Thermometers are provided at the inlet and outlet of the cooling coil. The cooling water enters the coil through a rotameter and the water from the tank is continuously withdrawn from the bottom of the tank and circulated through the shell side of the heat exchanger. The impeller speed is measured electronically and can be adjusted using variable power supply.

Procedure

A wide variety of heat transfer experiments can be devised for this unit. Several studies of the major process parameters are outlined below.

Studies Using the Unbaffled Stirred Tank

Set a constant steam rate to the unit. Devise a set of experiments to study the impeller speed and the flowrate of water through the external heat exchanger. This set of process variables should be selected to keep the tank temperatures below 85° Celsius.

Studies Using the Baffled Stirred Tank

Measurements should be made at the same conditions as in the above section except with the baffles inserted in the system. *Note: these experiments can most easily be done in conjunction with those in the previous section.*

Unsteady-State Heat Transfer in an Unbaffled Tank

In this experiment, the unsteady-state heat transfer rate is measured as a function of time. To simplify the experiment, the recirculation pump and internal cooling coil are not used.

Make experiments at two impeller speeds. Run the experiments until the tank temperature approaches 85° C.

Data Analysis

A correlation from the literature is available for unbaffled tanks with an impeller somewhat similar to that used in the stirred tank experiments. The equations below give an expression for the heat transfer coefficient in terms of physical properties and in terms of the Nusselt number, Prandtl number, and Reynolds number. *Note: see the following section "Appendix" for definitions of Symbols (SI units).*

$$\frac{(\frac{hD_T}{k})}{(\frac{C_P\mu}{k})^{.33}(\frac{\mu}{\mu_w}^{.14})} = .36\left(\frac{D_I^2 n\rho}{\mu}\right)^{.67} \tag{3}$$

or

$$log\left[\frac{Nu_T}{Pr^{\frac{1}{3}}}\right] = log(.36) + .67log[Re_I] \tag{4}$$

Quantitative agreement may not be good due to the different impeller but the correlation can be used to test qualitative agreement and the effect of process parameters.

For the steady-state experiment, Equation **(2)** can be used to calculate the heat transfer coefficient from the tank and condensing steam temperatures.

For the unsteady-steady state studies, a heat balance on the tank fluid gives

$$AU(t_1 - t_3) = mC_p \left[\frac{dt_3}{d\theta}\right] \tag{5}$$

where the term on the left refers to the rate of heat addition to the tank and the term on the right refers to the increase in internal energy of the fluid. The heat capacity of the fluid is C_p and θ is the time. It is assumed that the system is adiabatic and no heat is lost to the surroundings. The derivative can be determined from the slope of t_3 versus θ. This allows calculation of the overall heat transfer coefficients.

In each experiment, a heat balance should be presented as a check on the accuracy of the experimental measurements.

Appendix A

List of *SI* Symbols [units]

Symbol	**Description**
Q_{13}	heat transfer from steam to fluid in vessel [W]
Q_{hx}	heat removed by external heat exchanger [W]
Q_c	heat removed in cooling coils [W]
Q_{surr}	miscellaneous heat loss to surroundings [W]
h_i ; $h_i, wall$	heat transfer coefficient at inner vend wall $[\frac{W}{m^2K}]$
h_o ; $h_{o,wall}$	heat transfer coefficient at outer wall surface, steam side $[\frac{W}{m^2K}]$
m_{st}	steam flowrate [kg/s]
m_{hx}	recirculation rate [kg/s]
m_c	cooling water flow in coils [kg/s]
m_v	total fluid in vessel [kg]
D_I	impeller diameter [m]
D_T	inner tank diameter [m]
$D_{T,o}$	outer tank diameter [m]
L_T	vertical length of tank [m]
A	heat transfer surface area $[m^2]$

References

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